Mayank Raj

Contact Information

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Education

California Institute of Technology — Pasadena, CA, USA
*Ph.D. in **Applied and Computational Mathematics*** — 2022 – 2027
- Advisor: Prof. Andrew M. Stuart
- GPA: 4.1

Indian Institute of Technology Madras — Chennai, India
*M.Tech. in **Data Science*** — 2020 – 2021
*B.Tech. in **Mechanical Engineering*** — 2016 – 2020
- Minor in **Applied Mathematics**
- GPA: 9.28/10

Employment History

Wells Fargo — Bangalore, India
Data Scientist — 2021 – 2022
- Developed an **automated OCR and regular-expression pipeline** to detect, extract, and structure data from unstructured PDF documents.
- Deployed the pipeline on the firm's **parallel computing** cluster, automating the end-to-end workflow from PDF ingestion and data extraction to structured-data generation.

Internships

Tata Research Design and Development Center — Pune, India
Research Intern in Machine Learning — Winter, 2020
- Developed a **machine-learning-driven framework for multi-objective optimization** of product designs, integrating CFD simulations with surrogate modeling.
- Built a **machine learning surrogate** model to rapidly predict separation factor and pressure drop, enabling efficient exploration of the particle-classifier design space.
- Performed **multi-objective optimization** to identify Pareto-optimal designs balancing separation performance against pressure drop.

RWTH Aachen University — Aachen, Germany
Research Intern in Computational Solid Mechanics — Summer, 2019
- Modeled the 3D brick-and-mortar microstructure of nacre using HyperMesh, representing the composite architecture of brittle calcium carbonate and ductile organic matrix.
- Conducted LS-DYNA simulations to characterize the mechanical properties and investigate the effect of material composition on the toughness of nacre-based composites.
- Demonstrated that increasing the organic-matrix fraction enhances composite toughness but results in a corresponding reduction in overall strength.

Ph.D. Research Background

- **Modeled non-Markovian dynamics** of viscoplastic materials by engineering a **custom recurrent neural network (RNN)** trained on sample load and response trajectories.

- ❑ **Enforced dynamic stability** by constraining the architecture to be convex.
- ❑ Implemented custom training pipelines **utilizing backpropagation, adjoint methods,** and **stochastic differential equations** to generate synthetic training trajectories.

Publications

A full list of publications on Google Scholar.

[3] **Mayank Raj** et al. "**A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables**". *Journal of the Mechanics and Physics of Solids* 217 (2026), p. 106807. ISSN: 0022-5096. DOI: https://doi.org/10.1016/j.jmps.2026.106807.

[2] Harkirat Singh, **Mayank Raj**, and Kaushik Bhattacharya. "Effective behavior of heterogeneous media governed by strain gradient elasticity". *Journal of the Mechanics and Physics of Solids* 212 (2026), p. 106583. ISSN: 0022-5096. DOI: https://doi.org/10.1016/j.jmps.2026.106583.

[1] **Mayank Raj**, Pramod Kumbhar, and Ratna Kumar Annabattula. "**Physics-informed neural networks for solving thermo-mechanics problems of functionally graded material**" (2022). arXiv: 2111.10751 [cs.CE].

Awards & Achievements

- ❑ The Bell Family Fellowship 2022
- ❑ Mitacs Globalink Scholarship 2020
- ❑ UGC-DAAD Research Scholarship 2019
- ❑ Top 0.15% in JEE-Advanced 2016

Relevant Coursework

Probability, Statistics & Stochastic Processes; Stochastic Differential Equations; Linear Algebra; Optimization; Deep Learning

Tools

PyTorch, Numpy, Pandas, Scipy, Scikit Learn